

DAVIDE CORTINOVIS

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[Google Scholar](#)

EDUCATION

Center for Brain/Mind Sciences, University of Trento

PhD in Cognitive and Brain Sciences

Rovereto, Italy

2021 - April 2026

My research is at the intersection of cognitive and computational neuroscience, focusing on two topics:

- **Brain organization:** I collect and analyze brain imaging data (fMRI) from human participants to study how object-selective brain regions are spatially arranged and what principles drive that arrangement.

The *key contribution* was the identification of a large-scale organizing principle based on the action-related properties of objects — whether an object is something you act with (like a tool) or act on (like food) — which explains the location and function of several category-selective brain regions.

- **Computational Modelling:** I test whether state-of-the-art artificial neural networks — especially *topographic models* meant to capture the spatial organization of the brain — can reproduce the spatial and functional organization observed in the human visual system.

The *key contribution* is that current models fail to capture the action-based organizational principles found in the brain, highlighting an important gap in how well these models truly reflect human visual processing.

Justus Liebig University Giessen

Visiting PhD

Giessen, Germany

2024

I spent three months in Martin Hebart's lab, where I analysed an fMRI dataset that I previously collected and ran a series of encoding and topographic modelling analyses to test category selectivity.

Center for Brain/Mind Sciences, University of Trento

Master in Cognitive Neuroscience. Final grade: 110 with honors

Rovereto, Italy

2019 - 2021

University of Leeds

Erasmus exchange

Leeds, United Kingdom

2018-2019

University of Bergamo

Bachelor of Psychology. Final grade: 110 with honors

Bergamo, Italy

2016 - 2019

PUBLICATIONS

1. **Cortinovis, D.**, Truong, N., Op de Beeck, H., & Bracci, S. (2025). Encoding models uncover fine-grained feature selectivity for bodies, hands and tools. *under review*. [preprint]
2. **Cortinovis, D.**, Truong, N., Op de Beeck, H., & Bracci, S. (2025). Investigating action topography in visual cortex and artificial neural networks. *Nature Communications*. [paper]
3. **Cortinovis, D.**, Peelen, M. V., & Bracci, S. (2025). Tool representations in human visual cortex. *Journal of Cognitive Neuroscience*, 37(3), 515-531. [paper]

CONFERENCE ABSTRACTS

1. **Cortinovis, D.**, Hebart, M., & Bracci, S. (2025). Comparing Object Selectivity in Visual Cortex and Topographic Deep Artificial Neural Networks. *Conference on Cognitive Computational Neuroscience*. [conference abstract]

2. **Cortinovis, D.**, Orlandi, G., van Campenhout, L., & Bracci, S. (2025). Object dimensions underlying food selectivity in visual cortex. *Journal of Vision*, 25(9), 1910-1910.
3. Bracci, S., **Cortinovis, D.**, & Garnuto, E. (2025). The Effect of Scene Clutter on Visual Representations. *Journal of Vision*, 25(9), 2061-2061.
4. **Cortinovis, D.**, Op de Beeck, H., & Bracci, S. (2023). The role of action-related properties in shaping the object space in the biological and artificial brain. *Journal of Vision*, 23(9), 4868-4868.
5. **Cortinovis, D.**, Op de Beeck, H., & Bracci, S. (2021). The organization of body-parts representations in Deep Convolutional Neural Networks. *Perception*, 50(1_SUPPL), 123-123.

POSTER AND TALK PRESENTATIONS

1. Object dimensions underlying food selectivity in visual cortex and topographic artificial neural networks. Poster and **Talk** presentation at Concepts, Action, Objects (CAOs) Workshop, Rovereto, Italy (May 2026)
2. Action-related organization of object representations in human visual cortex and topographic models. **Invited Talk** at Kalanit Grill-Spector lab, Stanford University, Stanford, USA (February 2026)
3. Object dimensions underlying food selectivity in visual cortex. **Talk** at Vision Sciences Society, St. Pete Beach, Florida, USA (May 2025). [link]
4. Category-selectivity, again: Hand-selective clusters in visual cortex support action-related behavioural goals. **Talk** at Organization for Human Brain Mapping (OHBM), Seoul, South Korea (June 2024).
5. Topography and visual cortex in DANNs. **Invited Talk** at Katharina Dobs' lab, Giessen University, Giessen, Germany (June 2024)
6. The action and animacy feature spaces reveal a functional dissociation between lateral and ventral occipitotemporal cortex. Poster presentation at Concepts, Action, Objects (CAOs) Workshop, Rovereto, Italy (May 2024)
7. Category-selectivity, again: Hand-selective clusters in visual cortex support action-related behavioural goals. **Talk** at TEAP, Regensburg, Germany (March 2024).
8. Category-selectivity, again: Hand-selective clusters in visual cortex support action-related behavioural goals. Poster presentation at EWCN, Bressanone, Italy (January 2024).
9. Unveiling the Dimensions of Action Object Space: Insights from fMRI and DCNNs. Poster Presentation at Neuro-AI Talks, Osnabrueck, Germany (September 2023)
10. Unveiling the Dimensions of Action Object Space: Insights from fMRI and DCNNs. Poster presentation at Seeing and Acting Workshop (SAW), Coimbra, Portugal (September 2023)
11. The role of action-related properties in shaping the object space in the biological and artificial brain. Poster presentation at Concepts, Action, Objects (CAOs) Workshop, Rovereto, Italy (May 2023)
12. The role of action-related properties in shaping the object space in the biological and artificial brain. Poster presentation at ECVP, Nijmegen, Netherlands (August 2022).

ACADEMIA SERVICES

- Reviewer for *Imaging Neuroscience*, *Cortex*, *Journal of Vision*, *NeuroImage*, *Communications Psychology*

TEACHING AND MENTORSHIP

- Teaching assistant for the course Computational Modeling of Perception for the master program in Cognitive Science at the University of Trento (2024/2025 and 2025/2026).

- Teaching assistant for the course Research Design and Hands-on Statistics for the master program in Cognitive Science at the University of Trento (2022/2023).
- Supervised four Master's students through the full research cycle: experimental design, stimulus construction, behavioral and fMRI data collection, and data analysis. All students contributed to work that resulted in currently published or submitted research.

COMMUNITY ENGAGEMENT AND OPEN SCIENCE AND EQUITY PRACTICES

- **Participant recruitment:** Consistently prioritized gender balance and ethnic diversity in recruiting participants across all fMRI studies, in line with best practices for representative and inclusive neuroscience research.
- **Public engagement in neuroimaging:** During fMRI data collection sessions, regularly invited students and members of the public to learn about brain imaging — how data are collected, how the scanner works, and how to visualize their own brain activity. This hands-on outreach aimed to demystify neuroscience and make it accessible to people with little-to-no neuroscientific background.
- **Open Science:** For all published studies, data, stimulus sets, and analysis code have been made publicly available via GitHub ([davidecortinovis-droid.github.io](https://github.com/davidecortinovis-droid)) and OSF (<https://osf.io/ctmbx/>), supporting reproducibility and open access to research outputs.

AWARDS

- Award for best abstract, CAOs conference, Rovereto, Italy, May 2026
- Fully-funded scholarship for the Master program at the University of Trento, 2019-2021.

ADDITIONAL INFORMATION

Languages	Italian (native), English (C1), Spanish (B1), French (B1)
Technical skills	MATLAB, Python, fMRI (data collection and data analysis)